# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2023
## FULL TEST –X
## PAPER –1
## TEST DATE: 21-05-2023

**Time Allotted: 3 Hours** **Maximum Marks: 180**

**General Instructions:**

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

**Section – A *(01 –06, 19 – 24, 37 – 42):*** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

**Section – A *(07 – 10, 25 – 28, 43 – 46):*** This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

**Section – B *(11 – 18, 29 – 36, 47 – 54):*** This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**MARKING SCHEME**

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +4 | If only (all) the correct option(s) is (are) chosen; |
| Partial Marks | : | +3 | If all the four options are correct but ONLY three options are chosen; |
| Partial marks | : | +2 | if three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option; |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | –2 | In all other cases. |

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct option is chosen. |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | –1 | In all other cases. |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct numerical value is entered at the designated place; |
| Zero Marks | : | 0 | In all other cases. |



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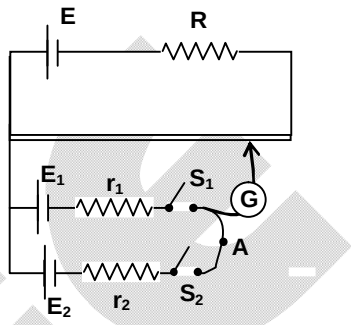
# Physics

## PART – I

### Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. Two batteries of emf $E_1$ and $E_2$ and internal resistance $r_1$ and $r_2$ respectively are connected across a potentiometer as shown. When only switch $S_1$ is closed, reading of potentiometer is $\ell_1$. When only switch $S_2$ is closed, reading of potentiometer is $\ell_2$. When both switches are closed, reading is $\ell_3$. Mark the **CORRECT** statement(s):



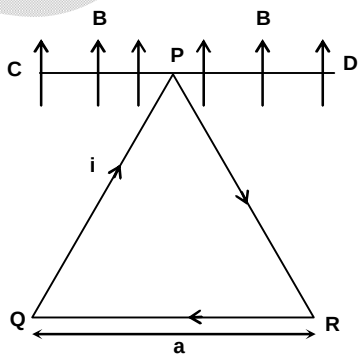

(A) $\ell_3 = \dfrac{\ell_1 + \ell_2}{2}$

(B) $\ell_1 < \ell_3 < \ell_2$ or $\ell_2 < \ell_3 < \ell_1$

(C) If another galvanometer is attached at point A in series and both switches are closed, it will show zero deflection.

(D) A change in primary circuit rheostat R changes all the readings of the potentiometer.

2. A loop PQR formed by three identical uniform conducting rods each of length 'a' is suspended from one of its vertices (P) so that it can rotate about a horizontal fixed smooth axis CD. Initially plane of loop is in vertical plane. A constant current 'i' is flowing in the loop. Total mass of the loop is 'm'. At t = 0, a uniform magnetic field of strength B directed vertically upwards is switched on. Acceleration due to gravity is 'g'.




(A) Minimum value of B so that the plane of the loop becomes horizontal (even for an instant) during its subsequent motion is $\dfrac{2mg}{3ia}$

(B) Minimum value of B so that the plane of the loop becomes horizontal (even for an instant) during its subsequent motion is $\dfrac{4mg}{3ia}$.

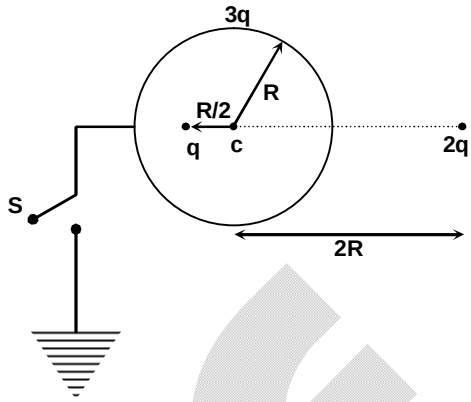
(C) The angle made by the plane of the loop with the vertical in equilibrium is $\tan^{-1}\left(\dfrac{3Bai}{2mg}\right)$

(D) The angle made by the plane of the loop with the vertical in equilibrium is $\tan^{-1}\left(\dfrac{3Bai}{4mg}\right)$



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3. A spherical shell of radius R is given charge 3q on its surface and a point charge q is placed at a distance R/2 from its centre C. Also there is a point charge 2q placed outside the shell at a distance 2R from its centre C as shown in the figure. Then:




(A) The magnitude of electric field at the centre C due to charges on the outer surface of shell is $\frac{Kq}{2R^2}$ before closing the switch S.

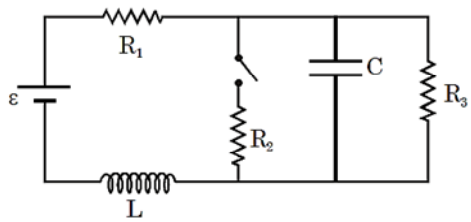
(B) The electric potential at the centre C due to charges on the outer surface of shell is $\left(\frac{-Kq}{R}\right)$ before closing the switch S.

(C) The electric potential at the centre C due to charges on the outer surface of shell is $\left(\frac{-Kq}{R}\right)$ after closing the switch S.

(D) Charge flown through the switch into earth after closing the switch S is 5q.

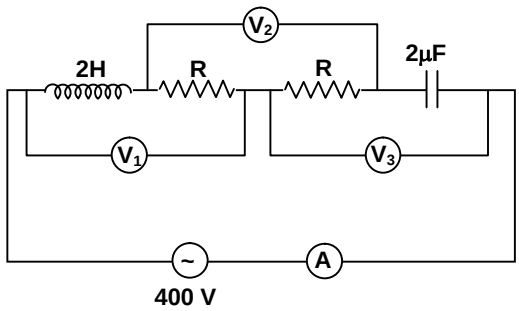
4. The circuit shown in the diagram is in the steady state with the switch open. When the switch is closed, which of the following will not change immediately after closing the switch?

(A) Potential difference across capacitor C.
(B) Current through the inductor L.
(C) Potential difference across resistance $R_1$.
(D) Current through the resistance $R_2$.




5. In the circuit as shown in the figure, the reading of voltmeters $V_1$ and $V_3$ are equal and the reading of the ammeter is 10A. Then choose the correct option(s).

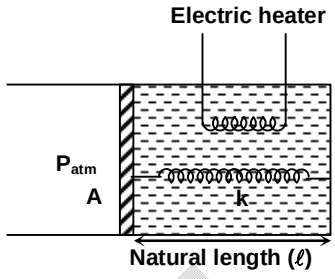
(A) $V_2 = 300$ V
(B) $V_2 = 400$ V
(C) $R = 10\Omega$
(D) $R = 20\Omega$






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6. One mole of an ideal monoatomic gas is enclosed in a thermally insulated cylinder fitted with a smooth adiabatic movable piston of area A with an electric heating arrangement as shown in the figure. If the piston is connected with a spring of stiffness $k = \frac{P_{atm}A}{\ell}$ and the gas is slowly heated such that the spring is stretched to twice its initial natural length ($\ell$) to store 25 J of energy, then choose the correct option(s).

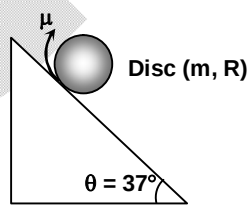



(A) Work done by the gas is 75 J
(B) Change in internal energy of the gas is 225 J
(C) Heat supplied by heater to the gas is 150 J
(D) Heat supplied by heater to the gas is 300 J

**Section – A (Maximum Marks: 12)**

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. Consider the following situation, where a disc of mass 2.5 kg is released from rest on a rough ($\mu$) inclined plane with angle of inclination, $\theta = 37°$ as shown in the figure. ($\mu$ = coefficient of friction between disc and inclined plane, g = 10 m/s$^2$)




Match the following two columns based on the above information.

| List –I | | List –II | |
|---|---|---|---|
| (I) | If $\mu$ = 0.4, then acceleration (in m/s$^2$) of centre of mass of disc will be | (P) | 2 |
| (II) | If $\mu$ = 0.3, then friction (in newton) acting on the disc will be | (Q) | 4.4 |
| (III) | If $\mu$ = 0.2, then acceleration (in m/s$^2$) of centre of mass of disc will be | (R) | 4 |
| (IV) | If $\mu$ = 0.1, then friction (in newton) acting on the disc will be | (S) | 5 |
| | | (T) | 3 |

Which one of the following options is correct ?
(A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ P
(B) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ S
(C) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ P
(D) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ S

8. Match the following list.

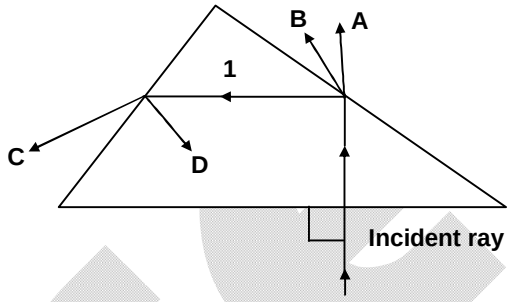
| List –I | | List –II | |
|---|---|---|---|
| (I) | Boltzmann constant | (P) | $[ML^2T^{-1}]$ |
| (II) | Coefficient of viscosity | (Q) | $[ML^{-1}T^{-1}]$ |
| (III) | Planck constant | (R) | $[MLT^{-3}K^{-1}]$ |
| (IV) | Thermal conductivity | (S) | $[ML^2T^{-2}K^{-1}]$ |
| | | (T) | $[ML^3T^{-2}K^{-2}]$ |



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Which one of the following options is correct ?
(A) I → S, II → Q, III → P, IV → R
(B) I → S, II → P, III → Q, IV → R
(C) I → R, II → Q, III → P, IV → S
(D) I → R, II → Q, III → S, IV → P

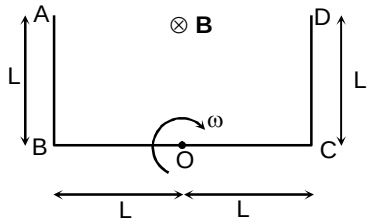
9. A white light ray is incident on the base of a glass prism, and it creates four refracted rays A, B, C and D. Match the refracted rays with the colours given (1 and D are rays due to total internal reflection):




| List –I | | List –II | |
|---|---|---|---|
| (I) | A | (P) | Red |
| (II) | B | (Q) | Green |
| (III) | C | (R) | Yellow |
| (IV) | D | (S) | Blue |
| | | (T) | White |

Which one of the following options is correct ?
(A) I → Q, II → R, III → P, IV → S
(B) I → P, II → R, III → Q, IV → S
(C) I → P, II → S, III → Q, IV → R
(D) I → Q, II → P, III → S, IV → R

10. A frame ABCD is rotating with an angular velocity ω about an axis passing through point O and perpendicular to the plane of paper as shown in the figure. A uniform magnetic field $\vec{B}$ is applied into the plane of the paper in the region as shown in the figure.




List-I describes the potential difference between the different points and List-II gives their respective values.

| List –I | | List –II | |
|---|---|---|---|
| (I) | Potential difference between O and A is | (P) | Zero |
| (II) | Potential difference between O and D is | (Q) | $\frac{B\omega L^2}{2}$ |
| (III) | Potential difference between C and D is | (R) | $\frac{B\omega L^2}{4}$ |
| (IV) | Potential difference between A and D is | (S) | $B\omega L^2$ |
| | | (T) | $2B\omega L^2$ |

Which one of the following options is correct ?
(A) I → S, II → S, III → Q, IV → P
(B) I → P, II → Q, III → R, IV → S
(C) I → S, II → T, III → P, IV → Q
(D) I → T, II → S, III → P, IV → R

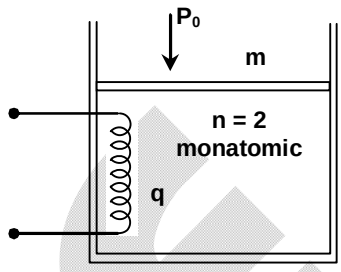


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## Section – B (Maximum Marks: 24)

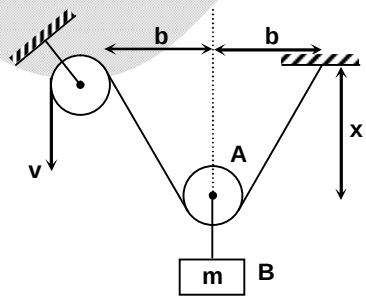
*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. Two moles of an ideal monoatomic gas is contained in a vertical cylinder of cross-sectional area A, as shown in the figure. The piston is frictionless and has a mass m. At a certain instant a heater starts supplying heat to the gas at a rate of q J/s. The velocity of the piston under isobaric condition is $\frac{10kq}{[p_0A+mg]}$. Find the value of k. (All the boundaries are thermally insulated. Atmospheric pressure is $p_0$)

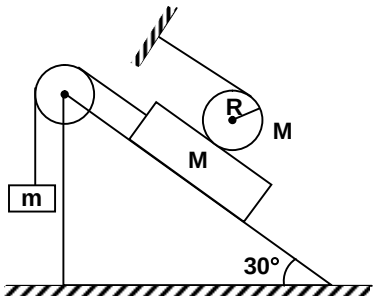



12. A balloon moves up vertically such that if a stone is thrown from it with a horizontal velocity $v_0$ relative to it the stone always hits the ground at a fixed point $\frac{2v_0^2}{g}$ horizontally away from it. The height of the balloon as a function of time is $\frac{100kv_0^2}{g}\left[1-e^{-gt/2v_0}\right]$. Find the value of k.

13. In the figure shown, one end of an ideal string is pulled down vertically with a constant velocity v. Assuming the pulleys to be massless and smooth. The tension in the string AB is $mg\left(1+\frac{b^2v^2k}{gx^3}\right)$. Find the value of k.




14. A plank of mass M is placed over smooth triangular wedge with angle of inclination 30°. Block of mass m is connected to the plank by massless string. A cylinder of mass M and radius R is now placed over the plank, such that one end of the string, wrapped over cylinder is fixed and parallel to incline. Coefficient of friction between cylinder and plank is sufficient for pure rolling of cylinder. If system is released from rest with condition that M = 4m. The acceleration of plank is $\frac{kg}{13}$ m/s$^2$. Find the value of k.



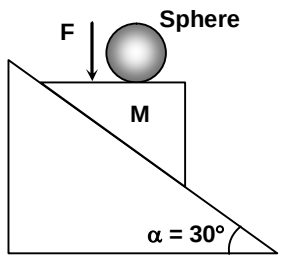

15. The smallest size area of a rectangular plane mirror, a person with a face 24 cm × 16 cm should use, which will enable him to see his whole face, is K cm$^2$. Find the value of K. (man is two eyed and separation between eyes is 8 cm)

16. A spherical raindrop initially of radius $r_0$ starts falling freely, receives water particle during its fall so that at every instant the rate of increase of volume is equal to $\mu$ times its surface area at that instant. The velocity of the raindrop after t seconds, assuming no air drag is $v=\frac{kg}{\mu}\left[r_0+\mu t-\frac{r_0^4}{(r_0+\mu t)^3}\right]$. Find the value of k.

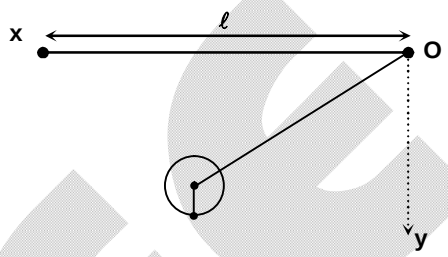


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17. A sphere of mass m is placed over a rough wedge of mass M which is placed on a smooth inclined plane as shown in the figure. Friction between the wedge and the incline is zero and that between the sphere and the wedge is sufficient to support pure rolling. The minimum value of force F required so that normal reaction between the sphere and the wedge becomes zero is KMg. Find the value of K.




18. A simple pendulum of length $\ell$ is suspended by a nail on a vertical wall at point O. Another nail is to be fixed on the wall such that if the pendulum is released from its initial horizontal position, it just completes a vertical circle round that nail as shown. The locus of the point where the second nail can be fixed is $K(x^2+y^2)=(3\ell-2y)^2$. Find the value of $\frac{K}{100}$.






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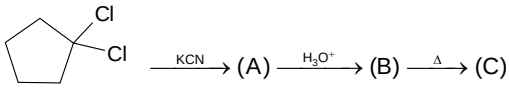
# Chemistry

## PART – II

### Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. Which compounds do not give Lassaigne test for nitrogen?
(A) $C_6H_5N_2Cl$ (B) $NH_2-CH_2-COOH$
(C) $NH_2OH$ (D) $CH_3CONH_2$

20. 


Identify the correct statement(s) about the above sequence of reactions:
(A) Compound (A) is formed through $S_N$ reaction.
(B) Compound (C) on reduction with $LiAlH_4$ forms a product which on dehydration gives cyclohexene.
(C) Compound (A) requires two moles of hydrogen for complete reduction.
(D) Compound (C) on Schmidt's reaction gives a product which reacts with $HNO_2$ to give (D) as major product. Compound (D) on dehydration gives cyclopentene.

21. Which of the following are condensation polymers?
(A) Glyptal (B) Nylon-66
(C) Dacron (D) PTFE

22. One mole of an ideal monoatomic gas at $27^oC$ expands adiabatically against constant external pressure of 1 atm from volume of 10 $dm^3$ to a volume of 20 $dm^3$.
(A) $\Delta U = -1013$ J
(B) $\Delta H = -1688$ J
(C) The final temperature achieved is 218.8 K
(D) The final temperature achieved is 105.2 K

23. $C_2H_4Cl_2 \xrightarrow{KCN} (B) \xrightarrow{H_3O^+} (C) \xrightarrow{\Delta} (D)$
(A)
The compound D has the characteristic(s)
(A) It is succinic anhydride if (A) is vic dihalide
(B) It is propionic acid if (A) is gem dihalide
(C) It is succinic anhydride if (A) is gem dihalide
(D) It is propionic acid if (A) is vic dihalide

24. Pyrophosphorus acid, $H_4P_2O_5$
(A) contains P in +5 oxidation state (B) is dibasic acid
(C) is strongly reducing in nature (D) contains one P – O – P bond

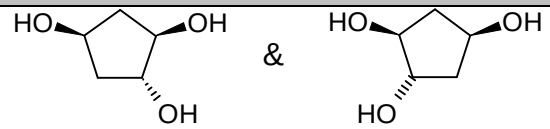
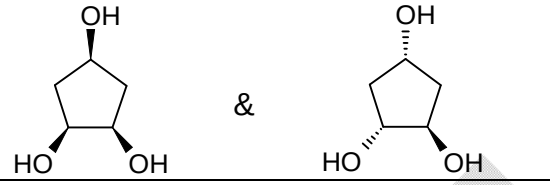


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**Section – A (Maximum Marks: 12)**

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the following:

| List – I | | List – II | |
|---|---|---|---|
| (I) |  | (P) | Identical |
| (II) |  | (Q) | Diasteromers |
| (III) |  | (R) | Enantiomers |
| (IV) |  | (S) | Positional isomer |
| | | (T) | Same magnitude of optical rotation |

(A) I→Q; II→P; III→R, T; IV→S
(B) I→R; II→Q; III→P; IV→ S
(C) I→R,T; II→P; III→Q; IV→S
(D) I→R; II→P, T; III→S; IV→Q

26. Match the following

| List – I (Solution) | | List – II pH (at 25°C) | |
|---|---|---|---|
| (I) | 50 mL of 0.5 M NaOH + 25 mL of 1 M $CH_3COOH(pK_a = 4.74)$ | (P) | 8.4 |
| (II) | 100 mL of 0.1 M aq. $HCO_3^-\left(pK_{a_1} = 6.4, pK_{a_2} = 10.4\right)$ | (Q) | 9.1 |
| (III) | 50 mL of 0.1 M HCl + 50 mL of 0.2 M $NH_4OH(pK_b = 4.74)$ | (R) | 9.3 |
| (IV) | 100 mL of 0.1 M $NH_4OH\left(K_b = 2\times10^{-5}\right)$ | (S) | 7 |
| | | (T) | 11.2 |

(A) I→S; II→P; III→R; IV→Q
(B) I→Q; II→P; III→S; IV→T
(C) I→S; II→R; III→P; IV→T
(D) I→Q; II→P; III→R; IV→T



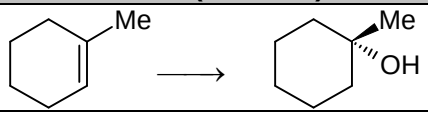
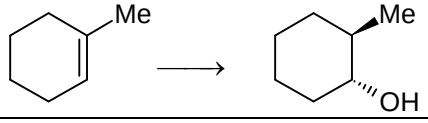
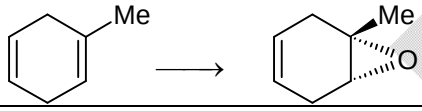
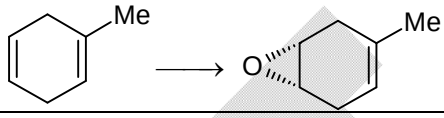
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27. Match the following

| List – I | | List – II | |
|---|---|---|---|
| (I) | $Ti(NO_3)_3$ | (P) | Lewis base |
| (II) | $(SiH_3)_3N$ | (Q) | Oxidizing agent |
| (III) | $SnCl_2$ | (R) | Reducing agent |
| (IV) | $(CH_3)_3N$ | (S) | $p\pi - p\pi$ overlap |
| | | (T) | $p\pi - d\pi$ overlap |

(A) I→R; II→S; III→Q; IV→P
(B) I→Q; II→S; III→P; IV→T
(C) I→Q; II→T; III→R; IV→P
(D) I→R; II→T; III→P; IV→Q

28. Match the following transformation with suitable reagent.

| List – I(Reaction) | | List – II(Reagent) | |
|---|---|---|---|
| (I) |  | (P) | mCPBA |
| (II) |  | (Q) | (i) $Hg(OAc)_2$, $H_2O$<br>(ii) $NaBH_4$ / $OH^-$ |
| (III) |  | (R) | $H_2O_2$ / $OH^-$ |
| (IV) |  | (S) | (i) $B_2H_6$, THF<br>(ii) $H_2O_2$ / $OH^-$ |
| | | (T) | $B_2H_6$ / $CH_3COOH$ |

(A) I→Q; II→S; III→P; IV→R
(B) I→Q; II→S; III→R; IV→P
(C) I→S; II→Q; III→P; IV→R
(D) I→R; II→S; III→Q; IV→P

**Section – B (Maximum Marks: 24)**

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. The heat of solution and the heat of formation of HCl(g) are – 76.5 kJ/mol and – 87.80 kJ/mol. Find the heat of formation of $Cl^-(aq.)$ ( in kJ/mol) (in magnitude)

30. 500 ml solution of 0.3 M $Al^{+3}$ is electrolysed with a current of 96.5 A. After electrolysis $Al^{+3}$ remaining is 0.2 M, find the duration of time (seconds) of electrolysis

31. The freezing point 0.08 molal $KHCO_3$ is $-0.372^oC$. Calculate the percentage of $HCO_3^-$ ion that transfer $H^+$ ion to water. Assuming 100% ionization of $NaHCO_3$. $K_f$ of $H_2O$ = 1.86 K mol/lit

32. NaOH and $Na_2CO_3$ mixture require 15 ml of $\frac{N}{20}H_2SO_4$ solution using phenolphthalein as indicator. But the same amount of solution when titrated with methyl orange as an indicator 20 ml of same acid. Calculate amount of NaOH in milligram.



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33. A sample of Argon gas at 1 atm pressure and at $27^{o}C$ expands reversibly and adiabatically from 1.25 L to 10.0 litre. Calculate $\Delta H$ change fo this process, $C_V = 3$ cal / mole / K , R = 2 cal/mole/K

34. An element exists in the body-centred cubic structure whose edge length is $3\overset{o}{A}$. The density of element is 8 g/cc. The number of atoms in 100 gm of the element are $y \times 10^{23}$. The value of y is____

35. 10 gram mixture of CO and $CO_2$ reacted with excess of $I_2O_5$ and liberated $I_2$ required 100 ml of $\frac{M}{2}Na_2S_2O_3$. Find the % of CO in mixture.

36. A mixture of acetaldehyde and ethanol weighing 0.484 gram when heated with Fehling solution gave 1.2 g red ppt. What is the percentage of acetaldehyde in mixture (At. wt. of Cu = 63.8 gm)



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# Mathematics

## PART – III

### Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. If a, b, c, d are positive rational numbers such that $a \pm b$ and $c \pm i\sqrt{d}$ satisfy the equation $\left(\frac{x}{2}-3\right)^4 = 1-\left(\frac{x}{2}-4\right)^4$, then
(A) a – b + c – d = number of zeroes at the end of 27!
(B) none of the number a, b, c, d is composite
(C) a + b + c + d is even
(D) a + b + c + d is odd

38. For the set of given equations / inequations $(\log_2 x)(\log_y 2)+1=0$; sin x cos y = 1 – cos x sin y and x + y < 8, which of the following statements are CORRECT?
(A) Number of pairs in the form of pair (x, y) satisfying the equation / inequations is 2
(B) Number of pairs in the form of pair (x, y) satisfying the equation / inequation is 4
(C) Sum of real values of x satisfying the equations / inequations is the fundamental period of the function $f(x)=\sin\left(\frac{4x}{5}\right)$
(D) Sum of real values of y satisfying the equations / inequations is the fundamental period of $f(x)=\sin\frac{4x}{5}$

39. The locus of the image of the focus of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ with respect to any of the tangent to the ellipse is
(A) $(x + 4)^2 + y^2 = 100$ (B) $(x + 2)^2 + y^2 = 50$
(C) $(x – 4)^2 + y^2 = 100$ (D) $(x – 2)^2 + y^2 = 50$

40. A straight line touches the rectangular hyperbola $9x^2 – 9y^2 = 8$ and the parabola $y^2 = 32x$ the equation of the line is
(A) 9x + 3y – 8 = ∞ (B) 9x – 3y + 8 = 0
(C) 9x + 3y + 8 = 0 (D) 9x – 3y – 8 = 0

41. Let f : (0, π) → R be a twice differentiable function such that $\lim_{t\to x}\frac{f(x)\sin t - f(t)\sin x}{t-x} = \sin^2 x$ for $\forall\, x \in (0, \pi)$. If $f\left(\frac{\pi}{6}\right)=\frac{\pi}{12}$, then which of the following statement(s) is/are TRUE?
(A) $f\left(\frac{\pi}{4}\right)=\frac{-\pi}{4\sqrt{2}}$ (B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$
(C) There exist $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) - f\left(\frac{\pi}{2}\right) = \pi$



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42. If $S = \frac{1}{1+n^2} + \frac{2}{2^2+n^2} + \frac{3}{3^2+n^2} + \ldots\ldots$. Further $S_1$ is the sum of first (n – 1) terms of the given series and $S_2$ is the sum of (n – 1) terms but starts from the 2nd term $\left(\text{i.e. } \frac{2}{2^2+n^2}\right)$ of the given series; n $\in$ N (set of natural numbers). Then which of the following is/are CORRECT?

(A) $S_1 > \frac{1}{2}\ln 2$ (B) $S_2 < \frac{1}{2}\ln 2$

(C) $S_2 > \frac{1}{2}\ln 2$ (D) $S_1 < \frac{1}{2}\ln 2$

**Section – A (Maximum Marks: 12)**

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Consider the curves $2|z - 2a| = z + \bar{z}$ and $|z - 4a| = 3a$ where $z = x + iy$

| List-I | | List-II | |
|---|---|---|---|
| (I) | If $z_1$ is the point of intersection of common tangents (not parallel to imaginary axis), then $\frac{\lvert z_1\rvert}{a}$ is | (P) | A prime number |
| (II) | If A is the area of triangle formed by chord of contact and common tangents from $z_1$ (used in 1st part), then $\frac{4A}{\sqrt{3}a^2}$ is (for $\lvert z - 4a\rvert = 3a$) | (Q) | Perfect square |
| (III) | Number of common points in both curves is | (R) | Perfect cube |
| (IV) | If equation of chord of contact from $z_1$ (used in part i) to the curve $2\lvert z - 2a\rvert = z + \bar{z}$ is $\frac{z+\bar{z}}{2} = ka$, then k is | (S) | Multiple of 10 |
| | | (T) | Multiple of 5 |

The correct option is:

(A) (I) → (P); (II) → (R); (III) → (T); (IV) → (Q)

(B) (I) → (P); (II) → (Q); (III) → (T); (IV) → (R)

(C) (I) → (P); (II) → (R); (III) → (P); (IV) → (Q)

(D) (I) → (P); (II) → (Q); (III) → (P); (IV) → (R)

44. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = \begin{bmatrix} a_{11}(n) & a_{12}(n) \\ a_{21}(n) & a_{22}(n) \end{bmatrix}$. If $\lim_{n\to\infty} \frac{a_{12}(n)}{a_{22}(n)} = \ell$ where $\ell^2 = \sqrt{a} + \sqrt{b}$ (a, b $\in$ N)

| List-I | | List-II | |
|---|---|---|---|
| (I) | a | (P) | Prime number |
| (II) | b | (Q) | Perfect square |
| (III) | a + b | (R) | Perfect cube |
| (IV) | $a^2 + b^2$ | (S) | Multiple of 5 |
| | | (T) | Multiple of 10 |



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The correct option is:
(A) (I) → (Q); (II) → (R); (III) → (T); (IV) → (S)
(B) (I) → (Q); (II) → (R); (III) → (P); (IV) → (S)
(C) (I) → (P); (II) → (S); (III) → (R); (IV) → (T)
(D) (I) → (P); (II) → (R); (III) → (T); (IV) → (S)

45. Match the following List-I with List-II

| List-I | | List-II | |
|---|---|---|---|
| (I) | If $ax + by - 5 = 0$ is the equation of the chord of the circle $(x-3)^2 + (y-4)^2 = 4$ which passes through (2, 3) and at the greatest distance from the centre of the circle, then | (P) | $a + b = 1$ |
| (II) | Let O be the origin and P be a variable point on the circle $x^2 + y^2 + 2x + 2y = 0$. If the locus of mid-point of OP is $x^2 + y^2 + 2ax + 2by = 0$, then | (Q) | $a + b = 2$ |
| (III) | If (a, b) be coordinates of the centre of the smallest circle which cuts the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x + 12y + 52 = 0$ orthogonally, then | (R) | $a^2 + b^2 = 2$ |
| (IV) | If a and b are the slopes of tangents which are drawn to the circle $x^2 + y^2 - 6\sqrt{3}x - 6y + 27 = 0$ from the origin, then | (S) | $a^2 + b^2 = 3$ |
| | | (T) | $a^2 - b^2 = 1$ |

The correct option is:
(A) (I) → (Q, S); (II) → (P); (III) → (R); (IV) → (S)
(B) (I) → (Q, R); (II) → (P); (III) → (P); (IV) → (T)
(C) (I) → (Q, S); (II) → (P); (III) → (P); (IV) → (R)
(D) (I) → (Q, R); (II) → (P); (III) → (P); (IV) → (S)

46. Match the following List-I with List-II

| List-I | | List-II | |
|---|---|---|---|
| (I) | If $x \in \left[0, \frac{\pi}{2}\right]$, $\int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\,dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\,dt = \frac{\pi}{k}$, then the value of k is | (P) | 2 |
| (II) | If $f(x) = \int_{e^{2x}}^{e^{3x}} \frac{t}{\ln t}dt \ \forall\, x \in R$, then $\left.\frac{d(f(x))}{d(\ln x)}\right\|_{x=\ln 2} = 24B$, then the value of B is | (Q) | 6 |
| (III) | $\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx = -\frac{\pi}{\alpha}\ln(\beta)$, then $2\alpha + \beta$ is | (R) | 3 |
| (IV) | Number of points of discontinuity for $y = \frac{1}{u^2 + u - 2}$ where $u = \frac{1}{x+1}$ is/are | (S) | 4 |
| | | (T) | |

The correct option is:
(A) (I) → (S); (II) → (Q); (III) → (R); (IV) → (S)
(B) (I) → (S); (II) → (Q); (III) → (R); (IV) → (T)
(C) (I) → (S); (II) → (P); (III) → (Q); (IV) → (R)
(D) (I) → (S); (II) → (P); (III) → (Q); (IV) → (S)



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**Section – B (Maximum Marks: 24)**

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. If S is the sum of all the possible integral values of x satisfying the equation
$20\left[\frac{x^3-5x^2+6x}{4}-{}^{x-1}C_5\right]-4\left[{}^{x-1}C_4-4\left({}^{x-1}C_3\right)\right]>0$,
then number of divisor of S is

48. If N is the number of ways in which 10 biscuits can be distributed to five beggars seating in a row such that no two consecutive beggars remain without biscuit, then $\frac{N}{100}$ is

49. Suppose that the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let $P_1$ and $P_2$ be two parabolas with a common vertex at (0, 0) and with foci P at $(f_1, 0)$ and $(3f_2, 0)$ respectively. Let $T_1$ be a tangent to $P_1$ which passes through $(3f_2, 0)$ and$T_2$ be a tangent to $P_2$ which passes through $(f_1, 0)$. If $m_1$ is the slope of $T_1$ and $m_2$ is the slope of $T_2$, then the value of $\left(\frac{2}{m_1^2}+3m_2^2\right)$ is

50. If the orthocentre of the triangle formed by the points $t_1$, $t_2$, $t_3$ on the parabola $y^2 = 4ax$ is the focus, the value of $\left|t_1t_2+t_2t_3+t_3t_1\right|$ is

51. The plane denoted by $P_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $P_2 : 5x + 3y + 10z = 25$. If the plane in its new position is denoted by P, and the distance of this plane from the origin is d, then the value of $\left[\frac{d}{10}\right]$ is (where [.] represents greatest integer function)

52. If the domain of the function f(x) which is defined as
$f(x)=\frac{1}{\sqrt{\sin(\cos x)}}+\sin^{-1}\left(\frac{2x}{\pi}\right)+\frac{1}{\{-x\}}+\frac{1}{\ln\left(1-\left[\tan\frac{x}{2}\right]-\left[-\tan\frac{x}{2}\right]\right)}$ is $x \in (a, b) - \{c, d, e\}$, then the value of (b + c + d + e) – a is equal to
(Where [.] represents greatest integer function and {x} = x – [x])

53. Let $f(x) = x^3 - 3x - 1$, then the number of distinct real solution of the equation f(f(x)) = 0 for $x \in (2, \infty)$

54. If f(x) is a real valued function such that $f(x + 1) = 1 + [2 - 3f(x) + 3f^2(x) - f^3(x)]^{1/3}$, where $f(x) \neq 0$ and $f(x + 2022) = \frac{2023}{\lambda}f(x)$, then $\lambda$ is equal to



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